Science practice booklet: Grade 8 expectations

Grade 8 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 8 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: MS-LS2-1, MS-LS2-2, MS-LS2-3, MS-LS2-4, MS-LS4-1, MS-LS4-2, MS-LS4-4

A worked example

In a mangrove ecosystem, algae are eaten by small fish and the small fish are eaten by herons. The algae capture 5000 units of energy from sunlight. About 10 percent of the energy is passed on at each step up the chain. How much energy reaches the herons, and where does the rest go?

1. Algae are the producers, small fish are the first consumers and herons are the second consumers.
2. Algae to small fish: 10 percent of 5000 = 500 units.
3. Small fish to herons: 10 percent of 500 = 50 units.
4. The rest is used by each organism for living (moving, growing, keeping warm) and is given off as heat, so it is not passed on.

About 50 units reach the herons; the rest is used for life processes or lost as heat.

Questions

1. Write the process of photosynthesis as a word equation.

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2. Salem's body cells release energy from food by cellular respiration. Which two substances do the cells use up?

   A. Carbon dioxide and water

   B. Glucose and oxygen

   C. Glucose and carbon dioxide

   D. Oxygen and water

3. Algae in a farm pond near Al Ain capture 8000 units of energy from sunlight. About 10 percent is passed on at each step. How much energy reaches the small fish that eat the algae?

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4. In a mangrove food chain, why are there usually only three to five steps?

   A. Producers cannot make food for more than three groups

   B. Energy is destroyed when an animal is eaten

   C. The animals at the top eat too much

   D. Only about 10 percent of the energy passes on at each step, so little is left

5. Corals in warm seas have tiny algae living inside them. The algae make food for the coral, and the coral gives the algae shelter. What kind of relationship is this?

   A. Mutualism, because both organisms benefit

   B. Competition, because both need the same space

   C. Predation, because the coral eats the algae

   D. Parasitism, because the algae take food from the coral

6. Two kinds of birds in a mangrove forest both eat the same small fish. Which type of interaction is this, and what may happen if the fish become fewer?

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7. An oryx reserve has a fixed area, a limited amount of grass and a limited amount of water. After some years the number of oryx stops growing. What is the best explanation?

   A. Oryx always stop having young after ten years

   B. Predators in the reserve eat every young oryx

   C. Food, water and space are limited, so the reserve can support only a certain number

   D. The Sun gives less energy to the grass each year

8. Why is an ecosystem with many different species usually more stable than one with only a few?

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9. The carbon atoms in a fallen mangrove leaf can later be part of a crab. Describe how matter is cycled in an ecosystem to make this possible.

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10. Fossils of sea shells are found high in the Hajar mountains. What does this evidence tell scientists about the place long ago?

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11. The arm bones of a human, the wing bones of a bat and the flipper bones of a whale have a similar pattern, though the limbs do different jobs. What do scientists conclude from this evidence?

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12. Some lizards in a sandy desert are light brown and some are dark. Birds find the dark lizards easier to see. After many generations, what is most likely to have happened to the group?

    A. All the lizards turn light brown during their lives to hide

    B. A larger share is light brown, because more light brown lizards survive and have young

    C. A larger share is dark, because dark lizards eat more

    D. Nothing changes, because lizards do not pass on their colour

Section 2: Earth and space

Standards practised: MS-ESS1-1, MS-ESS1-2, MS-ESS1-3, MS-ESS1-4

A worked example

Three layers of sedimentary rock lie flat, one on top of another and have not been disturbed. Layer X is at the top and has leaf fossils. Layer Y is in the middle and has shell fossils. Layer Z is at the bottom and has fish fossils. Which layer is the oldest, and which fossils are the youngest?

1. In undisturbed layers, new sediment is laid down on top of the older sediment.
2. So the bottom layer, Z, is the oldest and the top layer, X, is the youngest.
3. Fossils are the same age as the layer they are found in, so the fish fossils are the oldest and the leaf fossils are the youngest.

Layer Z is the oldest; the leaf fossils in layer X are the youngest.

Questions

1. What causes the seasons on Earth?

   A. Earth moves much closer to the Sun in summer and farther away in winter

   B. The Moon blocks some sunlight in winter

   C. The tilt of Earth's axis changes how directly sunlight reaches each part of Earth

   D. Earth spins faster in summer than in winter

2. Summer days in the UAE are long and hot. Explain why, using the Sun's position and the tilt of the Earth.

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3. Some people think it is hotter in summer because Earth is closer to the Sun. Is this correct? Give a reason.

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4. Why does the shape of the Moon seem to change through the month?

   A. We see different amounts of its lit half as it orbits Earth

   B. Earth's shadow covers a different part of it each night

   C. The Moon itself grows and shrinks

   D. Clouds cover different parts of it

5. People in the UAE watch a solar eclipse. Which order of the three objects is correct, starting from the Sun?

   A. Sun, Earth, Moon

   B. Moon, Sun, Earth

   C. Earth, Sun, Moon

   D. Sun, Moon, Earth

6. There is not a solar eclipse every month, even though the Moon passes between the Earth and the Sun each month. Why not?

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7. What force keeps the planets moving in orbits around the Sun, instead of travelling off in straight lines?

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8. The Hope Probe took about seven months to travel from Earth and enter orbit around Mars in February 2021. Which planet is Mars counting outwards from the Sun, and is it a rocky planet or a gas giant?

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9. Light travels at about 300 000 km per second. The Sun is about 150 000 000 km from Earth. About how long does sunlight take to reach Earth?

   A. 80 minutes

   B. About 8 minutes

   C. About 8 seconds

   D. About 8 hours

10. Put these in order from smallest to largest: the Milky Way galaxy, the Sun, Earth, the solar system.

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11. Hind looks at a cliff with horizontal layers of rock. A crack (a fault) cuts straight through all the layers. What does this tell her?

    A. The crack is older than the layers

    B. The crack and the layers formed at the same moment

    C. The layers are younger than the crack

    D. The crack is younger than the layers

12. Earth is about 4.6 billion years old, but people have lived on it for only a very small part of that time. Explain how rock layers and fossils help scientists build a timeline of Earth's history.

Section 3: Matter, forces and energy

Standards practised: MS-PS4-1, MS-PS4-2, MS-PS4-3, MS-PS2-2, MS-PS2-3

A worked example

A family car travels 210 km in 2.5 hours. What is its average speed? If it keeps the same average speed, how far does it travel in 4 hours?

1. Speed = distance ÷ time.
2. 210 km ÷ 2.5 hours = 84 km per hour.
3. Distance in 4 hours = speed × time = 84 × 4 = 336 km.

84 km per hour; 336 km in 4 hours.

Questions

1. Hamdan hits a drum harder and the sound gets louder. Which property of the sound wave has increased?

   A. Amplitude

   B. Frequency

   C. Speed in air

   D. Wavelength

2. A sound wave has a frequency of 500 hertz and a wavelength of 0.68 m. Use speed = frequency × wavelength to find the speed of the sound.

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3. Two astronauts stand close together outside their spacecraft but cannot hear each other speak. Why?

   A. Space is too cold for sound to form

   B. Sound changes into light waves in space

   C. Their helmets destroy all sound waves

   D. Space is almost empty, and sound needs matter to travel through

4. Layla shines a torch at a flat mirror. The ray hits the mirror at 30 degrees to the normal (the line at right angles to the mirror). At what angle does it reflect?

   A. 15 degrees

   B. 30 degrees

   C. 60 degrees

   D. 90 degrees

5. List these waves in order from the longest wavelength to the shortest: X-rays, visible light, radio waves, ultraviolet.

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6. Why do people in the UAE wear sunscreen and sunglasses when they spend a long time in the midday sun?

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7. A digital signal is made of a code of 0s and 1s. Give one reason why digital signals are used to send pictures and measurements from the Hope Probe to Earth.

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8. Name the energy change that happens in a solar panel, and say why solar parks suit the UAE.

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9. Two magnets are placed with their north poles facing each other. What do you feel when you try to push them together? Name one way to make an electromagnet stronger.

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10. A cyclist rides 36 km in 2 hours. A runner covers 10 km in 1 hour. Who has the greater average speed, and by how much?

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11. A metro train travels 120 km in 1.5 hours. What is its average speed?

    A. 180 km per hour

    B. 120 km per hour

    C. 80 km per hour

    D. 60 km per hour

12. Omar's family car starts from rest and reaches 20 m per second after 5 seconds. What is its acceleration?

    A. 100 m per second squared

    B. 15 m per second squared

    C. 4 m per second squared

    D. 25 m per second squared

13. A force of 3000 N pushes a car of mass 1000 kg. Use force = mass × acceleration to find the acceleration. Then explain why a passenger should wear a seatbelt when the car stops suddenly.

Answer guide for parents

Section 1: Living things

1. Plants take in carbon dioxide and water, use light energy, and make glucose and oxygen. As words: carbon dioxide + water, with light energy, makes glucose + oxygen.
2. B. Glucose and oxygen. Cells use up glucose from food and oxygen to release energy. If your child chose A, they gave the substances that are made, not the ones used up.
3. Only 10 percent passes on, so 10 percent of 8000 = 8000 ÷ 10 = 800 units reach the small fish.
4. D. Only about 10 percent of the energy passes on at each step, so little is left. About 10 percent of the energy passes on at each step, so after a few steps too little is left for another group. If your child chose B, they thought energy is destroyed, but it is used up or lost as heat.
5. A. Mutualism, because both organisms benefit. Both the coral and the algae gain something, so it is mutualism. If your child chose D, they thought the algae only take food, but the coral gets food from them too.
6. Competition, because both birds need the same food. If the fish become fewer, each bird gets less to eat, so their numbers may fall.
7. C. Food, water and space are limited, so the reserve can support only a certain number. of oryx. If your child chose A, they made up a rule about age, but the limit comes from the resources.
8. With many species there are many food sources and links, so if one species falls, others can still survive. The ecosystem keeps working.
9. Decomposers break down the dead leaf and return carbon to the soil and air. Plants take it in again as carbon dioxide and build it into leaves, and animals eat the leaves, so the atoms are used again.
10. Sea shells come from animals that lived in the sea and were buried in sediment under water, so the land was once covered by sea. Later the rock was lifted up into mountains.
11. The limbs have a similar bone pattern even though they do different jobs. This suggests the animals share a common ancestor and changed over a very long time.
12. B. A larger share is light brown, because more light brown lizards survive and have young. Birds eat the dark lizards more, so more light brown ones survive and have young with their colour. If your child chose A, they thought each lizard changes colour in its life, but colour is passed on in genes.

Section 2: Earth and space

1. C. The tilt of Earth's axis changes how directly sunlight reaches each part of Earth. Earth's axis is tilted, so sunlight hits each part more directly at some times of year than others. If your child chose A, they thought distance causes seasons, but Earth is not much closer in summer.
2. In summer our part of Earth tilts towards the Sun, so the Sun is higher and its light is more direct. The days also have more hours of daylight.
3. No. The tilt of Earth's axis causes the seasons. Earth is actually a little farther from the Sun in the northern summer.

4. A. We see different amounts of its lit half as it orbits Earth. The Moon always has one lit half, and as it orbits Earth we see different amounts of that half. If your child chose B, they mixed phases up with Earth's shadow, which only causes a lunar eclipse.
5. D. Sun, Moon, Earth. In a solar eclipse the Moon is in the middle, between the Sun and Earth. If your child chose A, they put Earth in the middle, which is the order for a lunar eclipse.
6. The Moon's orbit is tilted compared with Earth's orbit. So most months the Moon passes a little above or below the line between the Sun and Earth, and its shadow misses us.
7. The Sun's gravity pulls the planets inwards, which bends their path into an orbit. Gravity is stronger with more mass and weaker with more distance.
8. Counting outwards from the Sun: Mercury, Venus, Earth, Mars, so Mars is the fourth planet. It is a rocky planet, like the three planets before it.
9. B. About 8 minutes. Time = distance ÷ speed = 150 000 000 ÷ 300 000 = 500 seconds, about 8 minutes. If your child chose C, they mixed up seconds and minutes, since 500 seconds is about 8 minutes.
10. Earth, the Sun, the solar system, the Milky Way. Earth is smaller than the Sun, and the Sun sits inside the solar system, which sits inside the galaxy.
11. D. The crack is younger than the layers. The layers had to be there first for the crack to cut through them, so the crack is younger. If your child chose A, they thought the crack came first, but it cuts through the layers.
12. Layers show the order of events, with the oldest at the bottom. Fossils show which living things and places existed at each time, so together they build a timeline.

Section 3: Matter, forces and energy

1. A. Amplitude. A larger amplitude means more energy, which we hear as a louder sound. If your child chose B, they mixed up loudness with pitch, which is what frequency changes.
2. Speed = frequency × wavelength = 500 × 0.68 = 340 m per second.
3. D. Space is almost empty, and sound needs matter to travel through. Sound needs matter such as air to travel through, and space is almost empty. If your child chose A, they thought cold stops sound, but sound travels well in cold air.
4. B. 30 degrees. The angle of reflection equals the angle of incidence, so 30 degrees goes out as 30 degrees. If your child chose C, they used the angle to the mirror (90 take away 30 is 60) instead of the angle to the normal.
5. Radio waves, visible light, ultraviolet, X-rays. The wavelength gets shorter as the waves carry more energy.
6. The Sun gives out ultraviolet waves that carry enough energy to harm skin and eyes. Sunscreen and sunglasses block or absorb much of it.
7. A digital signal is only 0s and 1s, so it can be copied without losing detail and errors can be found and fixed. The picture arrives clearly even from very far away.
8. A solar panel changes light energy into electrical energy. The UAE has many hours of strong sunshine and lots of open land, so solar parks suit it.
9. Like poles repel, so the two north poles push apart. A stronger electromagnet needs more turns of wire, a larger current or an iron core.
10. Cyclist: 36 ÷ 2 = 18 km per hour. Runner: 10 ÷ 1 = 10 km per hour. So the cyclist is faster by 8 km per hour.

11. C. 80 km per hour. Speed = distance ÷ time = 120 ÷ 1.5 = 80 km per hour. If your child chose B, they gave the distance and forgot to divide by the time.
12. C. 4 m per second squared. Acceleration = change in speed ÷ time = 20 ÷ 5 = 4 m per second squared. If your child chose A, they multiplied 20 by 5 instead of dividing.
13. Acceleration = force ÷ mass = 3000 ÷ 1000 = 3 m per second squared. When the car stops, the passenger's body keeps moving forward (inertia), so the seatbelt gives the force that stops it safely.